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DM545/DM871 – Linear and integer programming

Exercise Sheet 8, Autumn 2026

Solution:

Included.

Exercises with the symbol $^+$ are to be done at home before the class. Exercises with the symbol * will be tackled in class and should be at least read at home. The remaining exercises are left for self training after the exercise class.

Exercise 1$^+$

Show that the maximum matching in arbitrary graphs is a special sort of set packing problem.

Solution:

The set packing problem asks to find a maximum size subset of the columns of a matrix in such a way that no row is covered more than once. The maximum matching asks to find the maximum size subset of the set of edges such that no vertex is covered more than once. Hence, in the matching problem the incidence matrix has the same role as in the matrix of the set packing. It is a special sort of set packing because the incidence matrix has exactly two 1s in each column.